Supplementary material - robustness checks

All quantities use the same common sample, the same passes and the same pass-bootstrap draws as the main analysis. dRMSE and dCentred RMSD are OSCAR minus CONTROL (negative: OSCAR closer to the altimeter). Thresholds are cumulative and overlapping, so intervals are pointwise.

Table S1. Bias versus bias-removed error by regime

Regime	U_t (m/s)	N	Passes	Bias CONTROL (m)	Bias OSCAR (m)	dRMSE [95% CI] (m)	dCentred RMSD [95% CI] (m)
opposing	0.0	4096	65	-0.065	-0.014	-0.037 [-0.058, -0.016]	-0.030 [-0.045, -0.013]
opposing	0.2	1086	40	-0.168	-0.038	-0.070 [-0.117, -0.023]	-0.031 [-0.064, 0.003]
opposing	0.4	336	17	-0.275	-0.034	-0.149 [-0.267, -0.043]	-0.055 [-0.127, -0.003]
opposing	0.6	94	9	-0.504	-0.167	-0.348 [-0.412, -0.223]	-0.125 [-0.195, -0.041]
following	0.0	11539	65	-0.020	-0.094	-0.010 [-0.028, 0.006]	-0.027 [-0.040, -0.014]
following	0.2	6173	60	-0.022	-0.132	-0.009 [-0.035, 0.016]	-0.040 [-0.056, -0.021]
following	0.4	3043	49	-0.028	-0.162	-0.001 [-0.040, 0.035]	-0.046 [-0.066, -0.023]
following	0.6	1498	34	-0.040	-0.188	0.037 [0.000, 0.071]	-0.030 [-0.055, -0.003]
following	0.8	527	16	-0.074	-0.244	0.099 [0.062, 0.150]	-0.031 [-0.052, -0.002]
following	1.0	134	6	-0.023	-0.239	0.172 [0.089, 0.251]	0.012 [-0.040, 0.036]
total	0.0	15635	67	-0.032	-0.073	-0.017 [-0.032, -0.005]	-0.026 [-0.039, -0.014]
total	0.2	11497	66	-0.047	-0.100	-0.015 [-0.032, 0.001]	-0.030 [-0.045, -0.014]
total	0.4	6179	65	-0.080	-0.145	-0.005 [-0.030, 0.016]	-0.033 [-0.054, -0.012]
total	0.6	3441	57	-0.082	-0.149	-0.002 [-0.029, 0.023]	-0.032 [-0.059, -0.006]
total	0.8	1969	49	-0.119	-0.191	0.017 [-0.011, 0.045]	-0.032 [-0.058, -0.008]
total	1.0	964	39	-0.089	-0.187	0.047 [0.016, 0.079]	-0.014 [-0.037, 0.005]

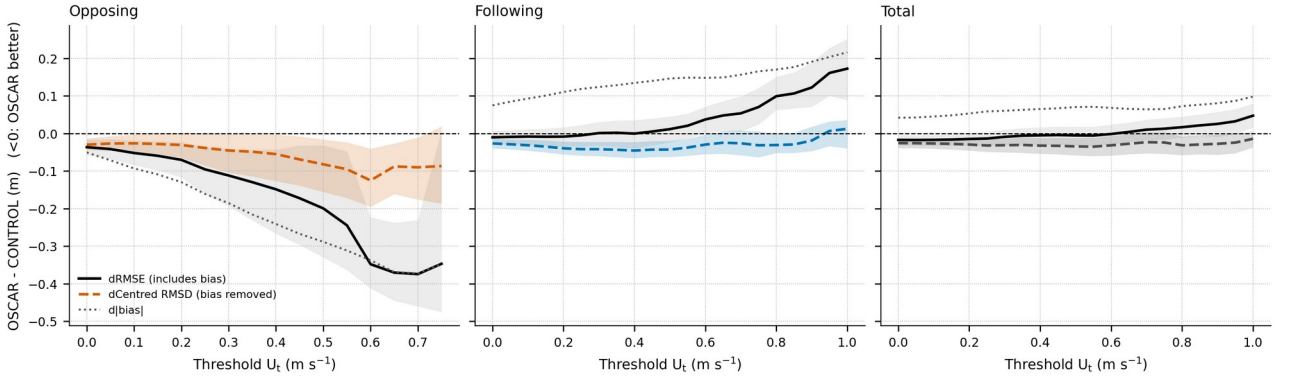


Figure S1. OSCAR minus CONTROL change in RMSE (black), centred RMSD with the mean bias removed (dashed, coloured) and in absolute bias (dotted), against threshold. Bands: 95% whole-pass bootstrap intervals.

Table S2. Consistency of the RMSE change across thresholds

Regime	Thresholds shown	U_t range shown (m/s)	OSCAR better (change < 0)	OSCAR worse (change > 0)	CI entirely < 0	CI entirely > 0	Smallest change (m)	Largest change (m)
opposing	16	0.00 to 0.75	16	0	15	0	-0.374	-0.037
following	21	0.00 to 1.00	7	14	0	9	-0.010	0.172
total	21	0.00 to 1.00	13	8	4	1	-0.018	0.047

Table S3. Bin-width sensitivity of the binned response

Bin width (m/s)	Bins shown	Spearman rho (centre vs median dHs)	Opposing bins with median dHs > 0	Following bins with median dHs < 0
0.050	33	-0.980	10 of 10	19 of 19
0.100	19	-0.996	7 of 7	10 of 10
0.200	10	-1.000	4 of 4	6 of 6

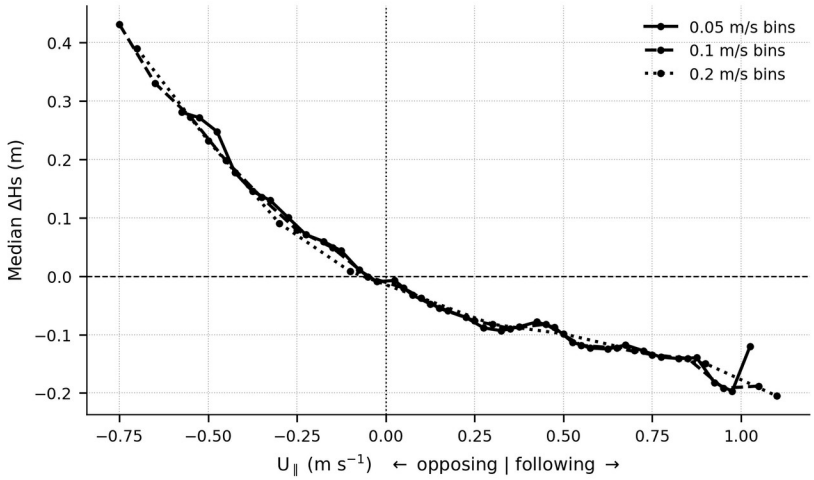


Figure S2. Median OSCAR minus CONTROL Hs against the along-wave current component for three bin widths.

Table S4. Leave-one-mission-out

Regime	U_t (m/s)	Mission dropped	N	Passes	dRMSE (m)	CI low	CI high	Enough data
opposing	0.200	(none dropped)	1086	40	-0.070	-0.117	-0.023	True
opposing	0.200	CryoSat-2	890	34	-0.085	-0.138	-0.026	True
opposing	0.200	HY-2B	1033	38	-0.071	-0.120	-0.023	True
opposing	0.200	HY-2C	926	35	-0.075	-0.123	-0.027	True
opposing	0.200	Jason-3	1011	38	-0.066	-0.116	-0.017	True
opposing	0.200	SARAL/AltiKa	980	37	-0.063	-0.110	-0.017	True
opposing	0.200	SWOT-nadir	972	35	-0.076	-0.128	-0.024	True
opposing	0.200	Sentinel-3A	991	34	-0.071	-0.120	-0.022	True
opposing	0.200	Sentinel-3B	936	35	-0.055	-0.101	-0.009	True
opposing	0.200	Sentinel-6A	949	34	-0.074	-0.122	-0.025	True
opposing	0.400	(none dropped)	336	17	-0.149	-0.267	-0.043	True
opposing	0.400	CryoSat-2	281	15	-0.189	-0.294	-0.060	True
opposing	0.400	HY-2B	308	15	-0.155	-0.282	-0.044	True
opposing	0.400	HY-2C	258	15	-0.172	-0.289	-0.058	True
opposing	0.400	Jason-3	312	16	-0.131	-0.252	-0.027	True
opposing	0.400	SARAL/AltiKa	290	14	-0.140	-0.273	-0.028	True
opposing	0.400	SWOT-nadir	327	16	-0.154	-0.279	-0.045	True
opposing	0.400	Sentinel-3A	312	16	-0.137	-0.263	-0.029	True
opposing	0.400	Sentinel-3B	286	14	-0.123	-0.266	-0.020	True
opposing	0.400	Sentinel-6A	314	15	-0.144	-0.272	-0.033	True
following	0.800	(none dropped)	527	16	0.099	0.062	0.150	True
following	0.800	CryoSat-2	453	14	0.102	0.062	0.161	True
following	0.800	HY-2B	516	15	0.099	0.062	0.152	True
following	0.800	HY-2C	406	12	0.128	0.090	0.174	True
following	0.800	Jason-3	450	14	0.097	0.058	0.157	True

following	0.800	SARAL/AltiKa	468	14	0.101	0.062	0.157	True
following	0.800	SWOT-nadir	460	14	0.097	0.058	0.155	True
following	0.800	Sentinel-3A	527	16	0.099	0.062	0.150	True
following	0.800	Sentinel-3B	438	14	0.080	0.054	0.116	True
following	0.800	Sentinel-6A	498	15	0.098	0.061	0.150	True
total	0.000	(none dropped)	15635	67	-0.017	-0.032	-0.005	True
total	0.000	CryoSat-2	13695	59	-0.019	-0.035	-0.005	True
total	0.000	HY-2B	15049	64	-0.020	-0.034	-0.007	True
total	0.000	HY-2C	13777	59	-0.018	-0.034	-0.004	True
total	0.000	Jason-3	13699	59	-0.016	-0.031	-0.002	True
total	0.000	SARAL/AltiKa	13881	60	-0.016	-0.031	-0.002	True
total	0.000	SWOT-nadir	13596	59	-0.018	-0.034	-0.004	True
total	0.000	Sentinel-3A	13992	58	-0.012	-0.022	-0.001	True
total	0.000	Sentinel-3B	13834	59	-0.018	-0.033	-0.004	True
total	0.000	Sentinel-6A	13557	59	-0.020	-0.035	-0.007	True